

Total No. of Questions : 8]

SEAT No. :

PC4850

[6353]-304

[Total No. of Pages : 2

T.E. (Artificial Intelligence and Machine Learning)

HONORS IN ARTIFICIAL INTELLIGENCE

(2019 Pattern) (Semester - II) (310303)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer four questions Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data if necessary.

- Q1)** a) Explain Bayesian inference using a suitable example. [9]
b) Explain Unification algorithm with suitable example. [8]

OR

- Q2)** a) Represent the following sentences into formulas in predicate logic. [9]
i) Apples are food.
ii) Chicken are food.
iii) John likes all kinds of food.
iv) Sue eats everything Bill eats.
v) Anything anyone eats and isn't killed by is food.
vi) Bill eats peanuts and is still alive.
b) Explain in detail Hidden Markov model with Example. [8]

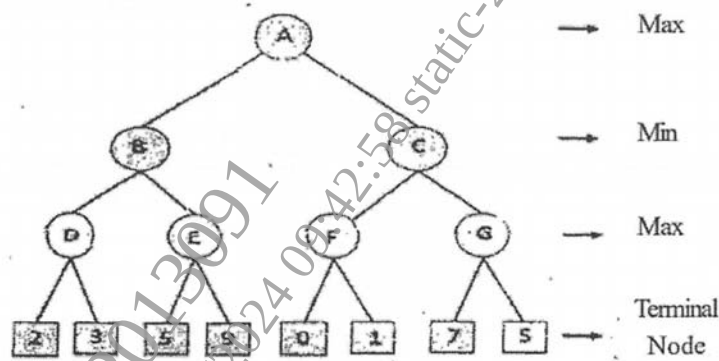
- Q3)** a) Explain how Support vector Machines are used for classification with suitable example. Also explain a real life problem that can be solved using SVM. [9]
b) What is decision tree? Explain how decision tree algorithm can be used in learning? [8]

OR

- Q4)** a) What is reinforcement learning? Explain how reinforcement learning can be applied for autonomous driving. [9]
b) With the help of an architecture diagram explain multilayer feed forward artificial neural network. [8]

P.T.O.

Q5) a) Solve the given game tree using min max algorithm **[9]**



b) Explain Games and Differentiate between Stochastic Games and Partial games with Example. **[9]**

OR

Q6) a) Explain Alpha-Beta Pruning with an example. What is the need of Alpha-Beta Pruning? **[9]**

b) Write a note on: **[9]**

- i) Mini-Max search for the tic-tac-toe game
- ii) State-of-the-art Game Programs

Q7) a) Explain computer vision as an expert system application with example of CV framework. **[9]**

b) Explain with example- Text Processing in Natural Language Processing. **[9]**

OR

Q8) a) Explain backward chaining algorithm for first order knowledge base. **[9]**

b) What is NLP? Explain all five phases of NLP. **[9]**

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